To overcome this issue with the memory, we present two naïve implementations, and three of the most widely used alternatives.

4.4.1 Unsorted Array: Fast Insertion, Slow Search

Even if you haven’t ever seen one of those paper dinosaurs (a dictionary), you ought to be familiar with at least print books, like this one (unless, of course, you bought the e-book version!).

Suppose you need to look for a specific word in the book, maybe a name, like Bloom. One option you have, is to go through the book starting from the first page, word by word, until you encounter it. If you need to find all the occurrences of the word Bloom, you will have to go through all the book from cover to back cover.

A book taken as a collection of words, in the order they are printed, is like an unsorted array; table 4.1 summarizes the performance of the main operations needed using this approach with unsorted arrays.

Table 4.1 Using unsorted arrays as dictionaries

Operation	Running Time	Extra Memory
`Creating the structure`	O(1)	No
	O(n)	No